

V3 Multi-Slice Bridge — Mini User Guide

V3 turns the bridge from a single receiver into a **Flex-style panadapter with multiple "slices"** on one captured window. One SDR, one LO — but several independent narrow receivers living inside the capture, each with its own VFO, mode, and output (WSJT-X over TCI, QMAP, the receiver popout, ...). The capture is wider; the slices roam inside it **digitally** (a per-slice DDC), so a second WSJT-X can chase signals without disturbing the first.

Defaults (v3.0): multi-slice is **ON by default** — the bridge comes up with Slice 0 + Slice 1 and the Slices window. Park mode defaults to **Band center** (DC-aware) and the demod to **USB**. To run plain single-slice, uncheck Settings → Radio → "Enable 2nd slice (roamer)" (restart to apply).

Slices, in order of importance

Slice 0 — the band-master (TCI 40001)

This is the boss. **Slice 0 is the ONLY slice that moves the SDR's LO / changes bands.** Drive it from your primary WSJT-X (the one on TCI **40001**, CAT 4535, and the bridge's WSJT-X UDP listener). When you change bands here, the whole capture window follows. Everything else is subordinate to Slice 0. It is also the only slice that can **transmit** (TX is a single shared front-end resource).

Slice 1 — the roamer (TCI 40002)

Digital-only. It tunes anywhere inside the captured window by shifting its DDC offset — it **never** moves the LO. Point a second WSJT-X at TCI **40002** and it decodes wherever you put it, with zero effect on Slice 0. It's **RX-only** (no TX — TX belongs to Slice 0) and it remembers its last frequency across restarts.

Rule of thumb: *Slice 0 picks the band; Slice 1 (and any further slices) live inside whatever window Slice 0 set.* If Slice 0 jumps to a band that doesn't contain Slice 1's frequency, Slice 1 is flagged out-of-range and muted.

Headline use case: **simultaneous FT8 + FT4** — Slice 0 on 14.074, Slice 1 on 14.080. Both fit one window, one LO, no second radio.

Slice 2 — the Receiver (the "Radio")

A third independent roamer — but instead of feeding WSJT-X, it's a **tunable multi-mode receiver played to your speakers**. Open it from **View → Receiver** (Ctrl+R). It has:

- its own **mode**: USB / LSB / CW / AM / NFM,
- a **tune** control (anywhere in the captured window),
- **volume**, and an **Output** device picker — defaults to your system speakers, separate from the WSJT-X VAC feed, so you can listen while WSJT-X decodes.

Like Slice 1 it's a digital-only roamer that never moves the LO. While it's open it shows as **slice 2 ("Receiver", roamer)** in the Slices window and as an **S2** marker on the panadapter. Use it to listen to SSB / a CW beacon / an FM repeater up the band while the digital slices decode at the bottom — one capture, one radio.

The Slices window and the Receiver are both on the **View menu** (Ctrl+I / Ctrl+R), so you can show/hide them without restarting.

Band-snap: what actually moves

The key V3 idea: **band-snap moves the LO park (the capture-window center), not your operating dial.** Your WSJT-X dial stays authoritative — the slice's digital DDC follows it. So:

- Tune **within** the current window → instant, pure-digital, **zero disruption** to the other slices.
- Tune to a **new band** (outside the window) → the LO **re-parks** (a brief retune) to re-center the window optimally for the new band.

That hysteresis means routine QSY never disturbs the roamer — only a real band change does.

Park modes — which to pick (`radio/park_mode`)

Where the LO parks the window center on a band change:

Mode	What it does	Use when
Off	Park on the dial (legacy V2). Window centered on your operating freq.	Simple single-slice, narrow capture.
EME	Snap to the EME sub-band centers (50.200 / 144.125 / 432.125 / 902.125 / 1296.125 ...).	EME — so QMAP sees the whole sub-band regardless of dial.
Band center ★	Per-band, DC-aware: parks the LO a small gap below the band's activity so the digital sub-band sits cleanly above the DC spike.	The default. General HF — keeps the DC spike off your signal.
Band lock	Locks to the band and sizes the capture width to cover it.	Wide-capable SDRs you want to see a whole band on.
Day/night block	Parks at the wide-block centers (night ~11.5 MHz covering 160–15 m, day ~19 MHz covering 30–10 m).	Multi-band capture in one window (RX-888 / Pluto-class).

LO placement & the DC spike

Where to put the LO depends on your SDR's **receive architecture**, not just the dial band.

ZIF / quadrature SDRs (SDRplay, Pluto, HackRF; and the RTL in normal-tuner / VHF mode) — the **DC spike sits AT the LO** (window center):

- **Narrow HF band** (e.g. 20 m): park the LO just **below the band edge** so the whole band sits one-sided *above* the spike — e.g. LO ~13.97 so DC < 14.000 and the 14.000–14.226 window covers FT8/FT4 above it. *Band-center* mode does this for you.
- **Wide allocation** (e.g. 2 m, 144–146): **center** the LO (~145.000) and accept the DC notch mid-band — 2 m activity (144.1 / 144.174) is narrow and off-center, so the spike falls in dead space.

Direct-sampling SDRs (RX-888; RTL HF Q-branch) — **no in-band DC spike** (DC is at true 0 Hz RF, far below HF), so **center freely**.

The real discriminator is the SDR's receive mode, not the dial frequency. A 28 MHz-IF transverter op reads ~28 MHz on the dial, but the RTL is back in normal-tuner (ZIF) mode there → the DC spike returns *and* it's an EME/QMAP use case → treat it like VHF (DC-aware park + 256 kHz feed), even though the dial says HF. The bridge keys this off the transverter/direct-sampling state, not the raw frequency.

VHF/UHF only: the DC-aware park and the 256 kHz QMAP/Linrad feed both matter at VHF/UHF (where you run EME and a ZIF tuner). On plain HF (direct sampling, no transverter) there's no in-band spike and no QMAP feed — just go as wide as the SDR/CPU/ADC bits allow for the waterfall + slices.

Quick setup (two slices)

1. **Multi-slice is on by default** — just **launch the bridge BEFORE WSJT-X** (TCI sessions don't survive a bridge restart, so restart WSJT-X after every bridge restart). (To disable it: Settings → Radio → uncheck "Enable 2nd slice".)
2. Slice 0 WSJT-X → **TCI 40001** (your main, band-master, TX-capable).
3. Slice 1 WSJT-X → **TCI 40002** (the roamer, RX-only). The 2nd-slice port is shown in Settings → Ports & Remote Control (it's always TCI port + 1).
4. Park mode defaults to **Band center**; switch to **EME** for EME, **Band lock** for a wide whole-band view.
5. **Change bands from Slice 0's WSJT-X** — the window follows, and Slice 1 keeps roaming inside it.
6. Want to *listen*? **View → Receiver** (Ctrl+R) opens the multi-mode receiver to your speakers (slice 2).

Transmit (TX)

TX is a **single shared front-end resource** — one TX chain, one LO — so two slices can never key at once.

- **Today:** only **Slice 0** (the band-master on TCI 40001) transmits. Slice 1 and the Receiver are **RX-only**, so there is no TX contention.
- **When multi-slice TX arrives:** keying will be **first-come, first-served, and never interrupted** — the first PTT to fire owns the TX chain; a second slice's PTT is rejected until the owner releases, and an in-progress transmission is never cut off. There is no priority or pre-emption: whoever keys first holds it.